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Lab4 – DS200.Q21

BÁO CÁO HÌNH KẾT QUẢ LAB 4

Bài 1 :

```
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$ python cau1.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/17 22:31:05 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/05/17 22:31:05 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/17 22:31:06 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
== orders ==
root
 |-- Order_ID: string (nullable = true)
 |-- Customer_Trx_ID: string (nullable = true)
 |-- Order_Status: string (nullable = true)
 |-- Order_Purchase_Timestamp: timestamp (nullable = true)
 |-- Order_Approved_At: timestamp (nullable = true)
 |-- Order_Delivered_Carrier_Date: timestamp (nullable = true)
 |-- Order_Delivered_Customer_Date: timestamp (nullable = true)
 |-- Order_Estimated_Delivery_Date: timestamp (nullable = true)
-----+-----
|Order_ID|Customer_Trx_ID|Order_Status|Order_Purchase_Timestamp|Order_Approved_At|Order_Delivered_Carrier_Date|Order_Delivered_Customer_Date|
-----+-----
|e481f51cbdc54678b7cc49136f2dea7|9ef432eb6251297304e76186b10a928d|delivered|2023-10-02 10:56:00|2023-10-02 11:07:00|2023-10-04 19:55:00|2023-10-10 21:25:00|
|2023-10-18 00:00:00|||
|53cdeb2fcbcb7dce0b6741e2150273451|b0830fb4747a6cd6d20dea0b8c802d7ef|delivered|2024-07-24 20:41:00|2024-07-26 03:24:00|2024-07-26 14:31:00|2024-08-07 15:27:00|
|2024-08-13 00:00:00|||
|47770eb9108c210c44946d9cf07ec65d|41ce2a54c0b03bf3443c3d931a367089|delivered|2024-08-08 08:38:00|2024-08-08 08:55:00|2024-08-08 13:50:00|2024-08-17 18:06:00|
|2024-09-04 00:00:00|||
|949d5b44dbf5de918fe9c16f97b45f8a|f88197465ea7920adcdbe37375364d82|delivered|2023-11-18 19:28:00|2023-11-18 19:45:00|2023-11-22 13:39:00|2023-12-02 00:28:00|
|2023-12-15 00:00:00|||
|ad21c59c0840e6c83a9ceb5573f8159|8ab97984e6daea8866dbdbcf4fb7aad2c|delivered|2024-02-13 21:18:00|2024-02-13 22:20:00|2024-02-14 19:46:00|2024-02-16 18:17:00|
|2024-02-26 00:00:00|||
-----+-----
only showing top 5 rows
== customers ==
root
 |-- Customer_Trx_ID: string (nullable = true)
 |-- Subscriber_ID: string (nullable = true)
 |-- Subscribe_Date: date (nullable = true)
 |-- First_Order_Date: date (nullable = true)
```

Bài 2 :

```
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$ python cau2.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/17 22:43:45 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/05/17 22:43:45 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/17 22:43:46 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Tổng số đơn hàng: 99441
Số lượng khách hàng: 99441
Số lượng người bán: 3095
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$
```

Bài 3 :

```
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$ python cau3.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/17 22:53:05 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/05/17 22:53:05 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/17 22:53:06 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

+-----+
|Customer_Country|Number_of_Orders|
+-----+
|Germany         |141754|
|France          |128488|
|Netherlands     |116229|
|Belgium         |15464 |
|Austria         |15043 |
|Switzerland     |3640  |
|United Kingdom  |3382  |
|Poland          |2139  |
|Czechia        |2034  |
|Italy           |2025  |
|Spain          |1651  |
|Portugal       |1336  |
|Sweden         |975   |
|Denmark        |905   |
|Serbia         |746   |
|Norway         |716   |
|Slovakia       |534   |
|Slovenia       |495   |
|Turkey         |485   |
|Greece         |412   |
+-----+

only showing top 20 rows
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$
```

Bài 4 :

```
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$ python3 cau4.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/18 09:27:59 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/05/18 09:27:59 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/18 09:28:00 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

KẾT QUẢ CÂU 4: Nhóm đơn hàng theo năm tăng dần, tháng giảm dần
+-----+
|year|month|num_rows|
+-----+
|2022|12   |1|
|2022|10   |324|
|2022|9     |4|
|2023|12   |5673|
|2023|11   |7544|
|2023|10   |4631|
|2023|9     |4285|
|2023|8     |4331|
|2023|7     |4026|
|2023|6     |3245|
|2023|5     |3700|
|2023|4     |2404|
|2023|3     |2682|
|2023|2     |1780|
|2023|1     |800|
|2024|10   |4|
|2024|9     |16|
|2024|8     |6512|
|2024|7     |6292|
|2024|6     |6167|
|2024|5     |6873|
|2024|4     |6939|
|2024|3     |7211|
|2024|2     |6728|
|2024|1     |7269|
+-----+
```

Câu 5 :

```
(pyspark-env) hdoop@lekhol-VMware-Virtual-Platform:~/DS200/Lab4$ python cau5.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/18 00:09:25 WARN Utils: Your hostname, lekhol-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/05/18 00:09:25 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/18 00:09:25 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

KẾT QUẢ CÂU 5: Review_Score
Điểm đánh giá trung bình (Review_Score 1-5): 4.09

Số lượng đánh giá theo từng mức Review_Score:
+-----+
|Review_Score|count|
+-----+
|1            |11424|
|2            |3151 |
|3            |8179 |
|4            |19141|
|5            |57328|
+-----+
```

Câu 6:

```
+-----+
|Product_Category_Name|Total_Sales|
+-----+
|Health_Beauty         |885191.1200000007|
|Watches_Gifts         |771986.7499999991|
|Bed_Bath_Table        |650794.6999999994|
|Sports_Leisure        |621999.3399999996|
|Computers_Accessories |594771.0400000003|
|Housewares            |491576.9600000005|
|Furniture_Decor       |476466.12999999983|
|Auto                  |404210.56999999995|
|Baby                  |299052.56000000006|
|Cool_Stuff            |273910.05000000005|
|Garden_Tools          |259068.31999999983|
|Telephony             |217452.12999999963|
|Perfumery             |204562.53999999986|
|Toys                  |200634.06999999998|
|Office_Furniture      |181745.73000000001|
|Stationery            |164743.85         |
|Pet_Shop              |152804.93999999992|
|Construction_Tools_Construction|141187.34         |
|Electronics           |134265.45000000007|
|Musical_Instruments   |121476.31000000001|
+-----+
only showing top 20 rows
(pyspark-env) hdoop@lekhol-VMware-Virtual-Platform:~/DS200/Lab4$
```

Câu 7:

```
(pyspark-env) hdoop@lekhol-VMware-Virtual-Platform:~/DS200/Lab4$ python cau7.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/18 10:17:22 WARN Utils: Your hostname, lekhol-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/05/18 10:17:22 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/18 10:17:23 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
San pham được mua nhiều nhất là:
+-----+
|Product_Category_Name|Number_of_Orders|
+-----+
|Bed_Bath_Table        |11115|
+-----+
only showing top 1 row
+-----+
|Product_Category_Name|Average_Review_Score|
+-----+
|Cds_Dvds_Musicals     |4.642857142857143|
|Fashion_Childrens_Clothes|4.5|
|Books_General_Interest|4.446265938069216|
|Construction_Tools_Tools|4.444444444444445|
|Flowers               |4.419354838709677|
|Books_Imported        |4.4|
|Books_Technical       |4.368421052631579|
|Food_Drink            |4.315412186379929|
|Luggage_Accessories   |4.315257352941177|
|Small_Appliances_Home_Oven_And_Coffee|4.302631578947368|
|Fashion_Sport         |4.258064516129032|
|Fashion_Shoes         |4.233716475095785|
|Food                  |4.218181818181818|
|Music                 |4.2105263157894735|
|Cine_Photo            |4.205479452054795|
|Stationery            |4.193057199800447|
|Pet_Shop              |4.185146982980918|
|Computers             |4.175|
|Home_Appliances       |4.172456575682302|
|Perfumery             |4.161940952937737|
+-----+
only showing top 20 rows
(pyspark-env) hdoop@lekhol-VMware-Virtual-Platform:~/DS200/Lab4$
```

Câu 10:

```
TypeError: exceptions must derive from BaseException
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$ python cau10.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/05/18 10:21:33 WARN Utils: Your hostname, lekhoi-VMware-Virtual-Platform, resolves to a loopback address: 127.0.1.1; using 192.168.139.128 instead (on interface ens33)
26/05/18 10:21:33 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/18 10:21:33 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Xếp hàng các Seller theo doanh thu:
+-----+-----+
|Seller_ID|Total_Sales|
+-----+-----+
|4869f7a5dfa277a7dca6462dcf3b52b2|240640.69999999999|
|7c67e1448b00f6e969d365cea6b010ab|239536.439999999994|
|53243585a1ddc2643021fd1853d8905|235856.68|
|4a3ca9315b744ce9f8e9374361493884|235539.959999999967|
|fa1c13f2614db75c4749c52fecda94|204084.73|
|da8622b14eb17ae2831f4ac5b9dab84a|185192.32000000015|
|7e93a43ef30c4f03f38b393420bc753a|182754.05000000005|
|1025f0e2d44d7041d6cf58b6550e0bfa|172860.69|
|7a67c85e85bb2ce8582c35f2203ad736|162648.38000000012|
|955fee9216a65b617aa5c0531780ce60|149992.60000000002|
|6560211a19b47992c3666cc44a7e94c0|151265.769999999967|
|1f50f920176fa81dab994f9023523100|142184.979999999986|
|46dc3b2cc0980fb8ec44634e21d2718e|139989.69|
|a1043baf4d71dff536d0c462352beb48|133745.25000000003|
|cc419e0650a3c5ba77189a1882b7556a|129957.409999999989|
|620c87c171fb2a6dd6e8bb4dec959fc6|128909.709999999999|
|5dceca129747e92ff8ef7a997dc4f8ca|125816.919999999998|
|7d13fca15225358621be4086e1eb0964|122261.02000000006|
|3d871de0142ce09b7081e2b9d1733cb1|117347.709999999999|
|edbf1f5e36e0c8cd84eb3c9b003e486d|83189.65|
+-----+-----+
only showing top 20 rows
So lượng đơn hàng của mỗi Seller:
+-----+-----+
|Seller_ID|Number_of_Orders|
+-----+-----+
|6560211a19b47992c3666cc44a7e94c0|2033|
|4a3ca9315b744ce9f8e9374361493884|1987|
|1f50f920176fa81dab994f9023523100|1931|
|cc419e0650a3c5ba77189a1882b7556a|1775|
|da8622b14eb17ae2831f4ac5b9dab84a|1551|
|955fee9216a65b617aa5c0531780ce60|1499|
|1025f0e2d44d7041d6cf58b6550e0bfa|1428|
|7c67e1448b00f6e969d365cea6b010ab|1364|
|ea8482cd71df3c1969d7b9473ff13abc|1203|
|7a67c85e85bb2ce8582c35f2203ad736|1171|
|4869f7a5dfa277a7dca6462dcf3b52b2|1156|
|3d871de0142ce09b7081e2b9d1733cb1|1147|
|8b321bb669392f5163d84c59e235e066|1018|
|cca3071cae9bb7d12640e9fbb2301306|830|
|620c87c171fb2a6dd6e8bb4dec959fc6|798|
|a1043baf4d71dff536d0c462352beb48|770|
|e9779976487b77c6d4ac45f75ec7afe9|750|
|f8db351d8c4c22c6835c19a46f01b0|724|
|d2374cbcb3ca4ab1086534108cc3ab7|631|
|391fc6631aebcf3004804e51b40bcf1e|613|
+-----+-----+
only showing top 20 rows
Traceback (most recent call last):
  File "/home/hdoop/DS200/Lab4/cau10.py", line 29, in <module>
    raise()
TypeError: exceptions must derive from BaseException
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$ python cau10.py
```

```
|edbf1f5e36e0c8cd84eb3c9b003e486d|83189.65|
+-----+-----+
only showing top 20 rows
So lượng đơn hàng của mỗi Seller:
+-----+-----+
|Seller_ID|Number_of_Orders|
+-----+-----+
|6560211a19b47992c3666cc44a7e94c0|2033|
|4a3ca9315b744ce9f8e9374361493884|1987|
|1f50f920176fa81dab994f9023523100|1931|
|cc419e0650a3c5ba77189a1882b7556a|1775|
|da8622b14eb17ae2831f4ac5b9dab84a|1551|
|955fee9216a65b617aa5c0531780ce60|1499|
|1025f0e2d44d7041d6cf58b6550e0bfa|1428|
|7c67e1448b00f6e969d365cea6b010ab|1364|
|ea8482cd71df3c1969d7b9473ff13abc|1203|
|7a67c85e85bb2ce8582c35f2203ad736|1171|
|4869f7a5dfa277a7dca6462dcf3b52b2|1156|
|3d871de0142ce09b7081e2b9d1733cb1|1147|
|8b321bb669392f5163d84c59e235e066|1018|
|cca3071cae9bb7d12640e9fbb2301306|830|
|620c87c171fb2a6dd6e8bb4dec959fc6|798|
|a1043baf4d71dff536d0c462352beb48|770|
|e9779976487b77c6d4ac45f75ec7afe9|750|
|f8db351d8c4c22c6835c19a46f01b0|724|
|d2374cbcb3ca4ab1086534108cc3ab7|631|
|391fc6631aebcf3004804e51b40bcf1e|613|
+-----+-----+
only showing top 20 rows
Traceback (most recent call last):
  File "/home/hdoop/DS200/Lab4/cau10.py", line 29, in <module>
    raise()
TypeError: exceptions must derive from BaseException
(pyspark-env) hdoop@lekhoi-VMware-Virtual-Platform:~/DS200/Lab4$ python cau10.py
```